Assignment 1 :~

Machine Learning

Lecture 8: OLS, Ridge, and Lasso Through One Worked Example

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Civil Engineering; Computer Science and Engineering

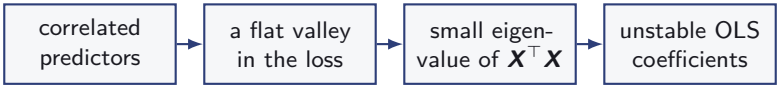
Semester I, 2026



What The Previous Lecture Established

Lasso / ridge

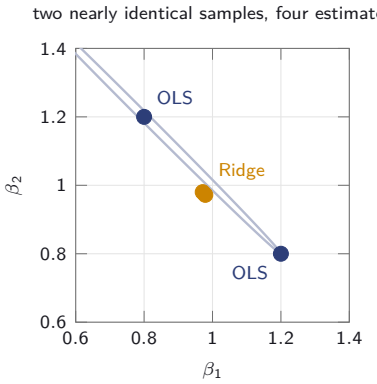
The board derivation followed this chain:



Summary

This lecture computes every object in that chain for one small dataset and *draws* each one — the valley, its two directions, what Ridge does to it — then repeats the calculation with Ridge and Lasso.

The Question For Today



When two predictors contain almost the same information, why can OLS assign very different coefficients to them, and how does regularization change the estimate?

OLS	**Ridge**	**Lasso**
best training fit	stable, smoothly shrunk	shrunk, exact zeros

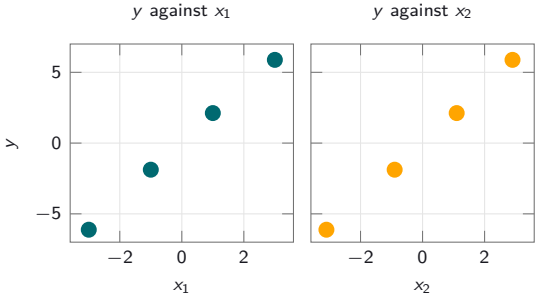
Question

The blue dots are OLS on two samples that differ by 0.08. The orange dots are Ridge on the same two. Which method should change least — and why?

One Dataset For The Entire Lecture

i	x_{i1}	x_{i2}	y_i
1	-3	-3.1	-6.12
2	-1	-0.9	-1.88
3	1	1.1	2.12
4	3	2.9	5.88

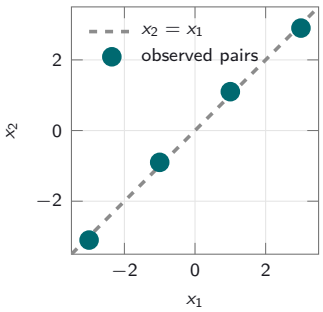
$$\boldsymbol{X} = \begin{bmatrix} -3 & -3.1 \\ -1 & -0.9 \\ 1 & 1.1 \\ 3 & 2.9 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} -6.12 \\ -1.88 \\ 2.12 \\ 5.88 \end{bmatrix}.$$



Meaning of one row

Row i is one observation: two measured features (x_{i1}, x_{i2}) and the response y_i. Either feature alone predicts y with slope about 2.

The Predictors Are Almost Duplicates



$y = \alpha_1 x_1 + \alpha_2 x_2$

$y = (\alpha_1 + \alpha_2)(x_1)$

$x_1 \approx x_2$

0.005

The two columns have

$$\|x_1\|_2 = 4.472, \qquad \|x_2\|_2 = 4.477.$$

They are centered and have almost the same scale.

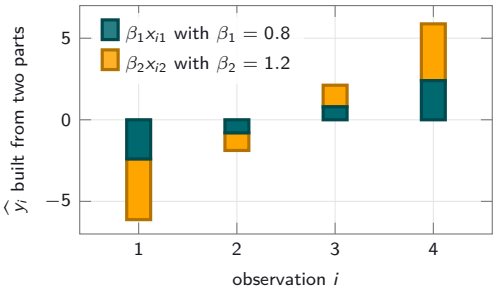
More importantly, every observed pair lies close to $x_2 = x_1$.

Insight

The model receives nearly the same information twice.

Interactive explorer: see how nearly parallel the two columns are ↗

The Prediction Rule



For observation i,

$$\widehat{y}_i = \beta_1 x_{i1} + \beta_2 x_{i2}, \qquad \widehat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{\beta}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}.$$

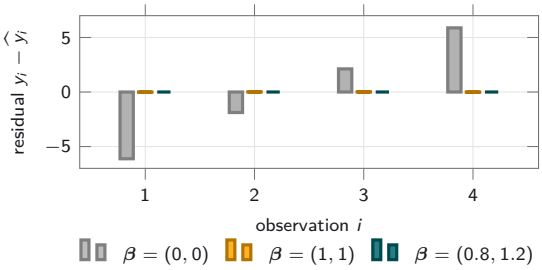
For the first row, $\widehat{y}_1 = -3\beta_1 - 3.1\beta_2$; at $\boldsymbol{\beta} = (0.8, 1.2)$ that is $-2.4 - 3.72 = -6.12$, the two stacked pieces of the first bar.

Summary

The features are centred, so the intercept is omitted to keep the arithmetic visible. In practice it is fitted and left unpenalized.

OLS Turns Prediction Errors Into One Number

$$J_{\mathrm{OLS}}(\beta) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\beta\|_2^2 = \frac{1}{2}\sum_{i=1}^{4}(y_i - \widehat{y}_i)^2.$$

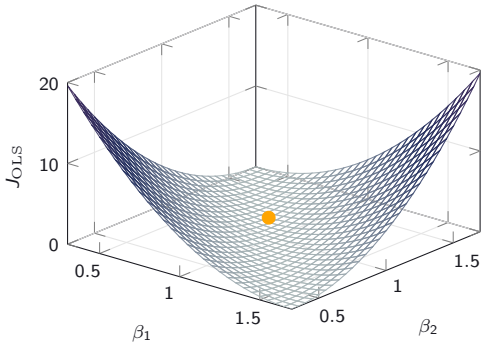


β	$J_{\mathrm{OLS}}(\beta)$
$(0, 0)$	40.0288
$(1, 1)$	0.0008
$(0.8, 1.2)$	0

Insight

Each bar is one residual; J is half the sum of their squares. OLS picks the pair whose bars are shortest overall — here all four vanish.

OLS Objective Expansion: J Is A Quadratic Surface



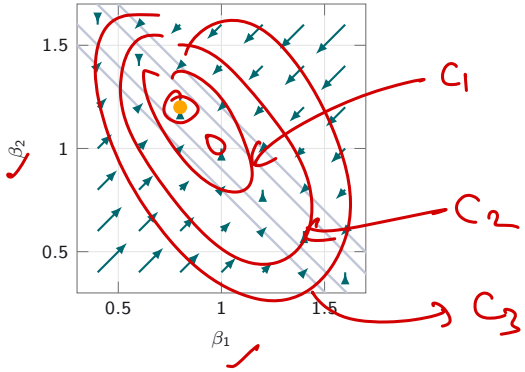
$$J_{\mathrm{OLS}}(\beta) = \tfrac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\beta)^\top(\boldsymbol{y} - \boldsymbol{X}\beta)$$
$$= \tfrac{1}{2}\boldsymbol{y}^\top\boldsymbol{y} - \beta^\top\boldsymbol{X}^\top\boldsymbol{y} + \tfrac{1}{2}\beta^\top\boldsymbol{X}^\top\boldsymbol{X}\beta,$$

using $\boldsymbol{y}^\top\boldsymbol{X}\beta = \beta^\top\boldsymbol{X}^\top\boldsymbol{y}$ (a scalar equals its transpose).

Insight

A constant, a linear term, and a quadratic term: J is a quadratic bowl over the coefficient plane. This one is a long trough with the orange minimum at its bottom — steep across, nearly flat along.

OLS Gradient: Every Arrow Points Across The Trough



Differentiate term by term:

$$\nabla\left(\tfrac{1}{2}\boldsymbol{y}^\top\boldsymbol{y}\right) = 0,$$
$$\nabla\left(-\boldsymbol{\beta}^\top\boldsymbol{X}^\top\boldsymbol{y}\right) = -\boldsymbol{X}^\top\boldsymbol{y},$$
$$\nabla\left(\tfrac{1}{2}\boldsymbol{\beta}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta}\right) = \boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta},$$

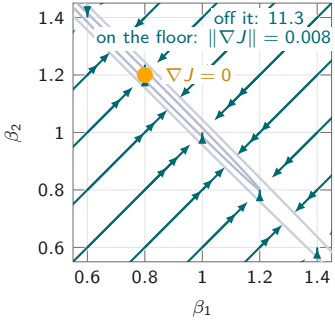
$$\boxed{\nabla J_{\mathrm{OLS}}(\boldsymbol{\beta}) = \boldsymbol{X}^\top(\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y}).}$$

Insight

The arrows are $-\nabla J$, drawn to scale: steepest descent at each point. Every one points across the trough, almost none along it, and they shrink to nothing at the floor (34 in the corner, 0.008 on the floor). The gradient sees the strong direction and is nearly blind to the weak one.

Interactive explorer: run gradient descent in the valley and watch it zigzag ↗

The OLS Normal Equation: Where The Arrows Vanish



At the minimum the gradient is zero:

$$\boldsymbol{X}^\top \boldsymbol{X}\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{X}^\top \boldsymbol{y} = 0 \implies \boxed{\boldsymbol{X}^\top \boldsymbol{X}\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{X}^\top \boldsymbol{y},}$$

and if $\boldsymbol{X}^\top \boldsymbol{X}$ is invertible, $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y}$.

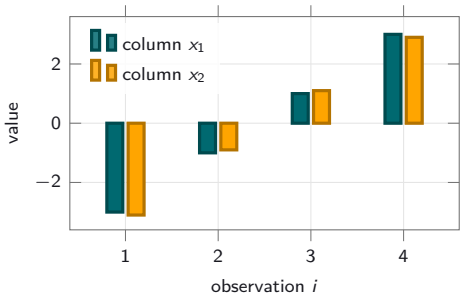
Two objects to compute

$\boldsymbol{X}^\top \boldsymbol{X}$ summarizes the relationships among the feature columns. $\boldsymbol{X}^\top \boldsymbol{y}$ summarizes how each feature aligns with the response.

Insight

The arrows vanish at the orange point, but they were already tiny all along the trough: equally far from the minimum, the gradient is 0.008 on the floor and 11.3 off it.

The Two Numerical Objects



The two columns almost coincide, so every dot product among them is almost the same number.

Feature geometry

$$\boldsymbol{X}^\top \boldsymbol{X} = \begin{bmatrix} x_1^\top x_1 & x_1^\top x_2 \\ x_2^\top x_1 & x_2^\top x_2 \end{bmatrix} = \begin{bmatrix} 20 & 20 \\ 20 & 20.04 \end{bmatrix}.$$

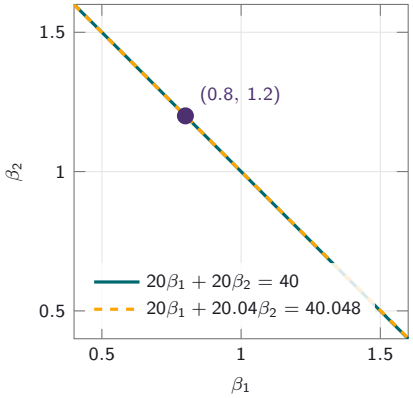
Feature–response alignment

$$\boldsymbol{X}^\top \boldsymbol{y} = \begin{bmatrix} x_1^\top \boldsymbol{y} \\ x_2^\top \boldsymbol{y} \end{bmatrix} = \begin{bmatrix} 40 \\ 40.048 \end{bmatrix}.$$

Warning

The Gram matrix has determinant $20(20.04) - 20^2 = 0.8$: positive, but tiny beside its entries.

The OLS Coefficients: Where Two Nearly Parallel Lines Cross



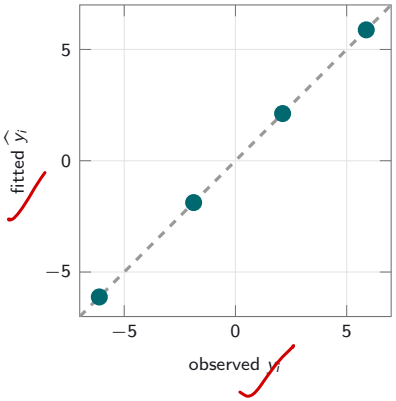
The normal equation is two lines in the (β_1, β_2) plane, and the solution is where they cross. Solving,

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \frac{1}{0.8}\begin{bmatrix} 20.04 & -20 \\ -20 & 20 \end{bmatrix}\begin{bmatrix} 40 \\ 40.048 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 1.2 \end{bmatrix}.$$

Insight

Slopes -1.000 and -0.998: at this scale they draw as one line. The crossing exists, but it is fixed by a difference of a thousandth — and the large entries in the inverse are the same warning in algebra.

Checking The OLS Fit



Multiply the coefficients back through the design matrix:

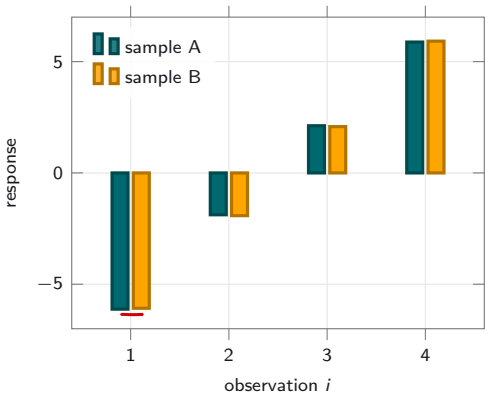
$$\boldsymbol{X}\widehat{\beta}_{\mathrm{OLS}} = \begin{bmatrix} -3 & -3.1 \\ -1 & -0.9 \\ 1 & 1.1 \\ 3 & 2.9 \end{bmatrix} \begin{bmatrix} 0.8 \\ 1.2 \end{bmatrix} = \begin{bmatrix} -6.12 \\ -1.88 \\ 2.12 \\ 5.88 \end{bmatrix} = \boldsymbol{y}.$$

Every fitted value sits on the diagonal: $\boldsymbol{y} - \boldsymbol{X}\widehat{\beta}_{\mathrm{OLS}} = 0$ and $J_{\mathrm{OLS}}(\widehat{\beta}_{\mathrm{OLS}}) = 0$.

Question

If OLS fits perfectly, what could still be wrong with the estimate?

A Slightly Different Training Sample



A fresh sample from the same setting gives slightly different responses:

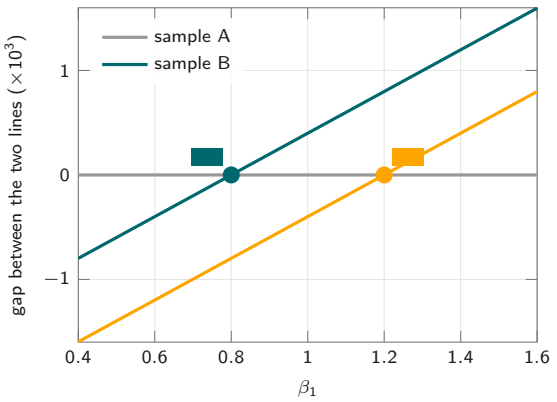
$$\boldsymbol{y}^{(A)} = \begin{bmatrix} -6.12 \\ -1.88 \\ 2.12 \\ 5.88 \end{bmatrix}, \quad \boldsymbol{y}^{(B)} = \begin{bmatrix} -6.08 \\ -1.92 \\ 2.08 \\ 5.92 \end{bmatrix},$$

so $\boldsymbol{y}^{(B)} - \boldsymbol{y}^{(A)} = (0.04, -0.04, -0.04, 0.04)^\top$, of length 0.08. The bars are indistinguishable by eye.

Summary

The design matrix stays fixed. Only the observed responses change, slightly.

OLS On The Slightly Changed Responses: The Crossing Jumps



Same Gram matrix, new right-hand side $\boldsymbol{X}^\top \boldsymbol{y}^{(B)} = (40, 40.032)$:

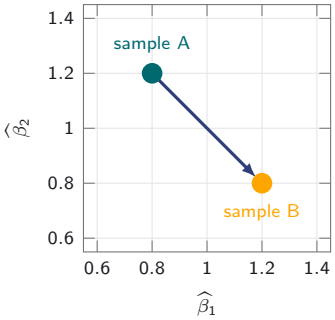
$$\widehat{\boldsymbol{\beta}}^{(B)}_{\mathrm{OLS}} = \frac{1}{0.8}\begin{bmatrix} 20.04 & -20 \\ -20 & 20 \end{bmatrix}\begin{bmatrix} 40 \\ 40.032 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}.$$

The plot shows the *gap* between the two lines, magnified a thousand times. The crossing is where the gap is zero. Its slope is 0.002, so shifting the gap by 0.0008 moves the crossing by 0.4.

Warning

Responses moved by 0.08; coefficients moved by $\sqrt{0.4^2 + 0.4^2} = 0.566$.

The OLS Coefficient Swap



$$\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}^{(A)} = \begin{bmatrix} 0.8 \\ 1.2 \end{bmatrix}, \qquad \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}^{(B)} = \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}.$$

Both coefficient pairs have the same sum:

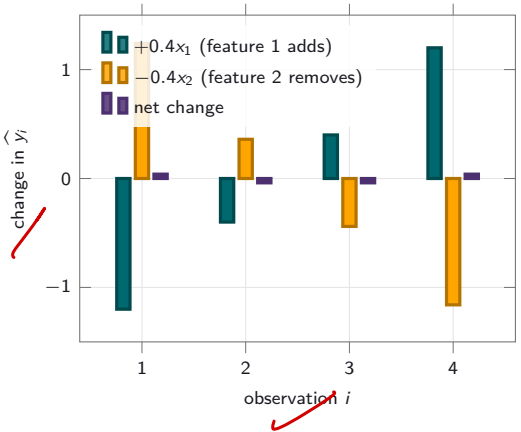
$$\widehat{\beta}_1 + \widehat{\beta}_2 = 2.$$

Because $x_1 \approx x_2$, the prediction depends mainly on this sum.

Insight

OLS changes how it divides credit between nearly duplicate predictors.

What Cancels When The Coefficients Swap?



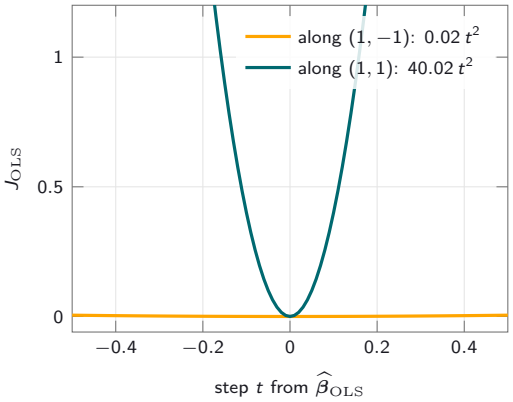
$$\widehat{\boldsymbol{y}}^{(B)} - \widehat{\boldsymbol{y}}^{(A)} = \underbrace{0.4x_1}_{\text{adds}} + \underbrace{(-0.4x_2)}_{\text{removes}}$$

$$= 0.4(x_1 - x_2) = \begin{bmatrix} 0.04 \\ -0.04 \\ -0.04 \\ 0.04 \end{bmatrix}.$$

Summary

The teal and orange bars are almost mirror images, so the purple bars — the actual change in the fitted values — are nearly nothing. **Cancellation happens in the prediction, not in the coefficients.**

Near Cancellation Makes OLS Coefficients Unstable



Move the coefficients by t in the swap direction:

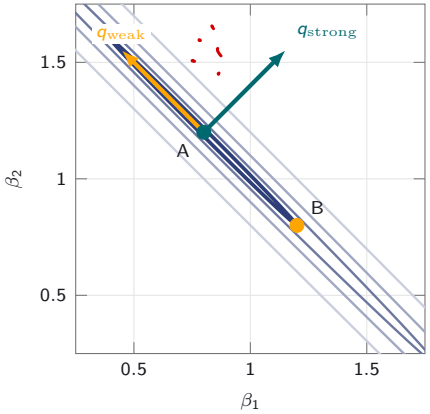
$$\beta_{\mathrm{new}} = \beta + t \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \boldsymbol{X}(\beta_{\mathrm{new}} - \beta) = t(x_1 - x_2) \approx 0,$$

so J rises by only $\frac{1}{2}t^2\|x_1 - x_2\|^2 = 0.02\,t^2$. Move by t along $(1,1)$ instead and it rises by $40.02\,t^2$ — two thousand times faster.

Warning

Tiny response noise can change which duplicate predictor gets the credit. Predictions barely move; the individual OLS coefficients are not reliable measures of importance.

The Loss Surface Is A Valley, Not A Bowl



Each curve joins coefficient pairs with the same training loss on sample A. The innermost one is $J_{\text{OLS}} = 0.0032$.

The curves are not circles. They are ellipses stretched **44.7 to 1**, long along $(1, -1)$ and thin across it: a valley with a nearly flat floor.

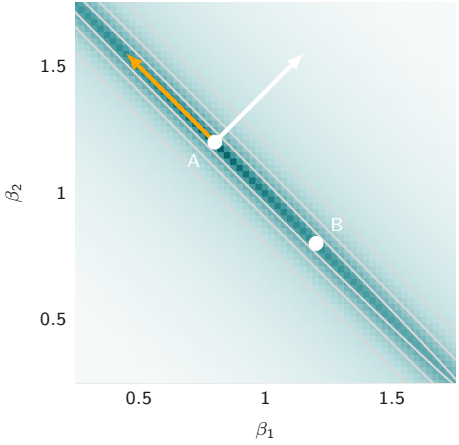
Sample B's minimum sits *on* the 0.0032 curve. The noise pushed the answer half a unit along the floor, and 0.0003 across it.

Insight

OLS is unstable in exactly one direction — along the floor.

Interactive explorer: move around the valley and read the loss ↗

The Same Surface As A Map: Dark Is Low



Colour is $\log_{10} J$: the darker, the lower. The dark band *is* the valley floor, and both OLS solutions sit inside it. Orange arrow: q_{weak}, along the band. White arrow: q_{strong}, off it.

Along the band, one unit of coefficient change costs $\frac{1}{2}\mu_{\text{weak}} = 0.01$ of loss. Off the band, the same unit costs $\frac{1}{2}\mu_{\text{strong}} = 20$.

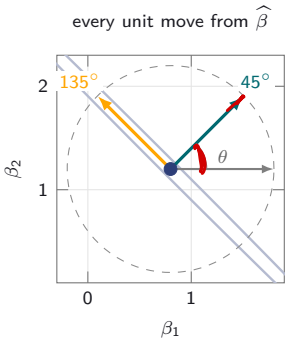
walk 0.25 from A	along	across
rise in J	0.0006	1.2506

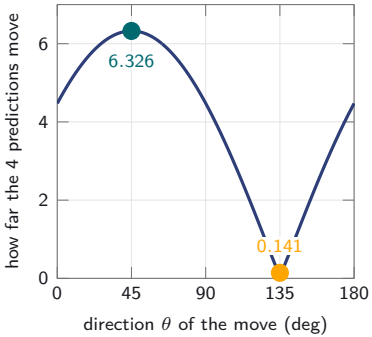
Insight

The ratio of the two rises is $\mu_{\text{strong}}/\mu_{\text{weak}} = 2002$, the condition number of $\boldsymbol{X}^\top \boldsymbol{X}$.

Interactive explorer: walk along the floor and across it ↗

Naming The Two Directions





Left: from $\widehat{\beta}$, move the coefficients by exactly one unit in every direction θ (the dashed circle). Right: for each direction, how much the four predictions change, $\|\boldsymbol{X}\boldsymbol{q}(\theta)\|$.

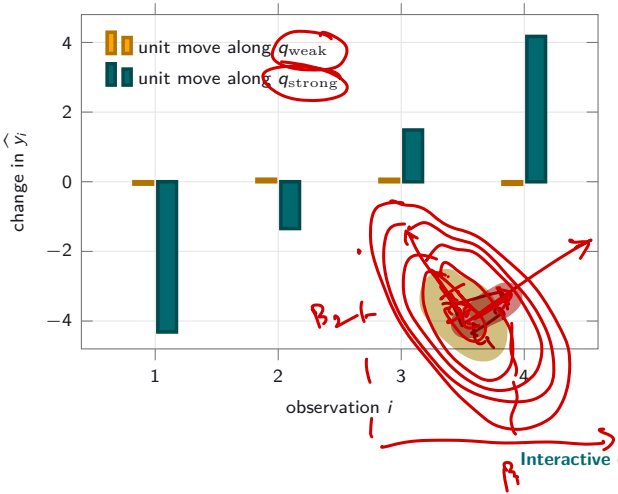
At 45° (both up together) they move 6.3. At 135° (one up, one down) they move 0.14. Those two directions are the eigenvectors of $\boldsymbol{X}^\top\boldsymbol{X}$, and $\|\boldsymbol{X}\boldsymbol{q}\|^2 = \mu$.

	q	μ
weak (135°)	$(-0.7075,\ 0.7068)$	0.01999
strong (45°)	$(0.7068,\ 0.7075)$	40.0200

Summary

"Weak" is not a judgement about a feature. It is the direction in *coefficient space* that the data barely constrain.

What A Weak Move Does To The Predictions



Move the coefficient vector by one unit and see what each observation's prediction does:

$$\boldsymbol{X}q_{\text{weak}} = \begin{bmatrix} -0.069 \\ 0.071 \\ 0.070 \\ -0.073 \end{bmatrix}, \quad \boldsymbol{X}q_{\text{strong}} = \begin{bmatrix} -4.31 \\ -1.34 \\ 1.49 \\ 4.17 \end{bmatrix}.$$

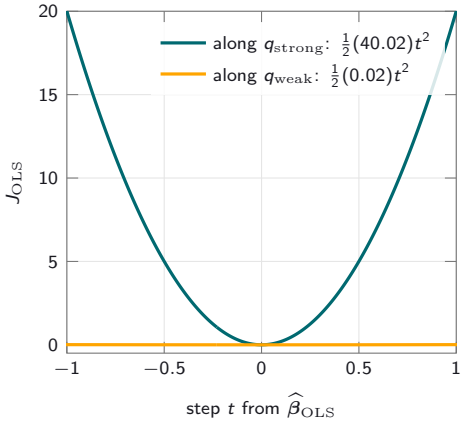
Lengths 0.141 and 6.326: a **45-fold** difference.

Insight

The weak direction is the coefficient change the predictions cannot see. That is why the data cannot pin it down — and why noise settles there.

Interactive explorer: step along each direction and watch the predictions ↗

Why The Small Eigenvalue Marks The Unstable Direction



For a unit eigenvector q_j,

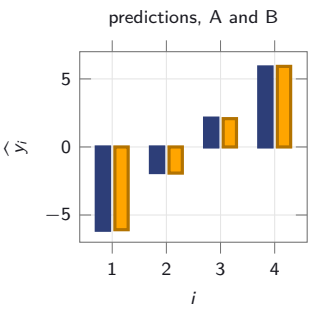
$$\begin{aligned} \boldsymbol{X}^\top \boldsymbol{X} q_j &= \mu_j q_j, \\ q_j^\top \boldsymbol{X}^\top \boldsymbol{X} q_j &= \mu_j \, q_j^\top q_j = \mu_j, \\ \|\boldsymbol{X} q_j\|_2^2 &= \mu_j; \end{aligned}$$

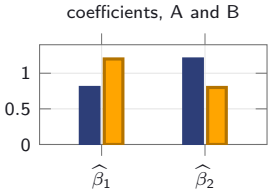
so stepping t along q_j raises the loss by $\frac{1}{2}\mu_j t^2$. Here $\|\boldsymbol{X} q_{\text{weak}}\|_2 = \sqrt{0.01999} \approx 0.141$ and $\|\boldsymbol{X} q_{\text{strong}}\|_2 = \sqrt{40.02} \approx 6.326$.

Insight

Curvature of the loss, size of the prediction change, and eigenvalue are the same number. The orange parabola is the trough floor; the teal one is its wall.

What The Weak Direction Means In Practice





For predictions

Almost nothing. Samples A and B predict the same $\hat{y}$ to within 0.04 — the top panel. Anyone who only needs $\hat{y}$ can stop worrying here.

For the coefficients

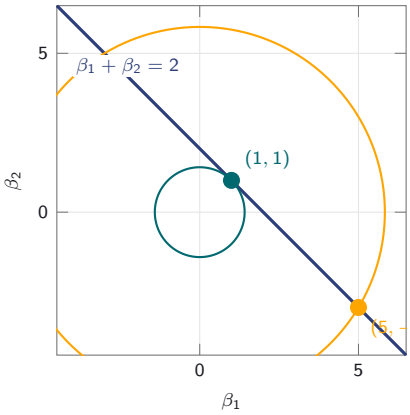
Everything — the bottom panel. Along the floor the data are silent, so 0.8 and 1.2 are equally supported. Neither is "the importance of x_1".

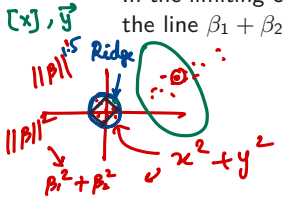
For what comes next

If the data will not choose a point on the floor, something else must. Ridge supplies it: a preference for the smallest coefficient vector.

Beyond two predictors: every small eigenvalue is another flat direction. Correlated features are the usual cause; too few observations the other.

Ridge Breaks The Tie Between Equivalent Fits





In the limiting case $x_2 = x_1$, $\widehat{\boldsymbol{y}} = (\beta_1 + \beta_2)x_1$: every point on the line $\beta_1 + \beta_2 = 2$ makes the same predictions.

(β_1, β_2)	sum	$\|\beta\|_2^2$
$(1, 1)$	2	2
$(5, -3)$	2	34
$(20, -18)$	2	724

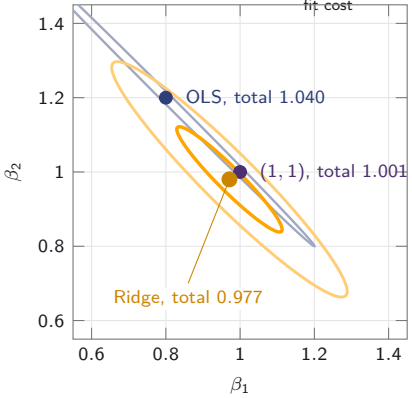
Ridge asks for the point on that line nearest the origin — the smallest circle that still touches it, at $(1, 1)$: $\beta_1^2 + \beta_2^2 = 2 + \frac{1}{2}(\beta_1 - \beta_2)^2$.

Summary

A tie-breaking preference, not evidence about the true coefficients.

The Ridge Preference In The Running Example

With $\alpha = 1$: $J_{\text{Ridge}}(\beta) = \underbrace{\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\beta\|_2^2}_{\text{fit cost}} + \underbrace{\frac{1}{2}\|\beta\|_2^2}_{\text{size cost}}$.



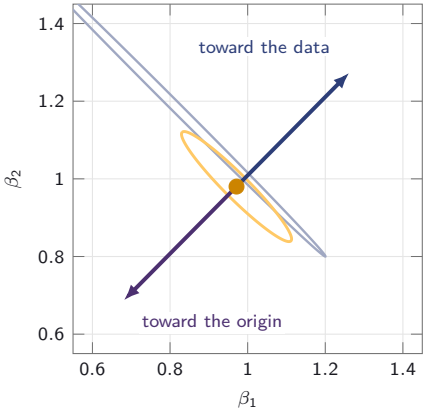
Candidate β	fit	size	total
$(0.8, 1.2)$, OLS	0	1.0400	1.0400
$(1, 1)$	0.0008	1.0000	1.0008
$(0.9713, 0.9801)$	0.0246	0.9520	**0.9766**
$(5, -3)$	0.3528	17.0000	17.3528

Blue: the OLS floor. Orange: contours of the *total* Ridge objective. Its lowest point is neither the best fit nor the smallest vector.

Insight

Ridge trades a little training error for a larger cut in unstable coefficient size. It minimizes the sum, not either column.

The Ridge Normal Equation: Two Pulls That Cancel



Differentiate both terms of J_{Ridge}:

$$\nabla J_{\text{Ridge}}(\beta) = \boldsymbol{X}^\top(\boldsymbol{X}\beta - \boldsymbol{y}) + \alpha\beta.$$

The fit term's descent direction $-\boldsymbol{X}^\top(\boldsymbol{X}\beta - \boldsymbol{y})$ pulls toward the data; the penalty's, $-\alpha\beta$, pulls toward the origin. At the solution they cancel exactly — here both have length 1.380:

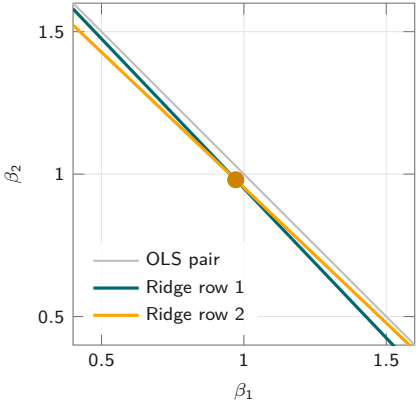
$$(\boldsymbol{X}^\top\boldsymbol{X} + \alpha I)\widehat{\beta}_{\text{Ridge}} = \boldsymbol{X}^\top\boldsymbol{y},$$

$$\boxed{\widehat{\beta}_{\text{Ridge}} = (\boldsymbol{X}^\top\boldsymbol{X} + \alpha I)^{-1}\boldsymbol{X}^\top\boldsymbol{y}.}$$

Summary

Algebraically $\boldsymbol{X}^\top\boldsymbol{X} \mapsto \boldsymbol{X}^\top\boldsymbol{X} + \alpha I$. Physically: a spring to the origin, with stiffness α.

The Ridge Matrix For $\alpha = 1$: The Lines Are No Longer Parallel



Add one to the diagonal of the Gram matrix:

$$\boldsymbol{X}^\top \boldsymbol{X} + \boldsymbol{I} = \begin{bmatrix} 21 & 20 \\ 20 & 21.04 \end{bmatrix}, \quad \det = 41.84.$$

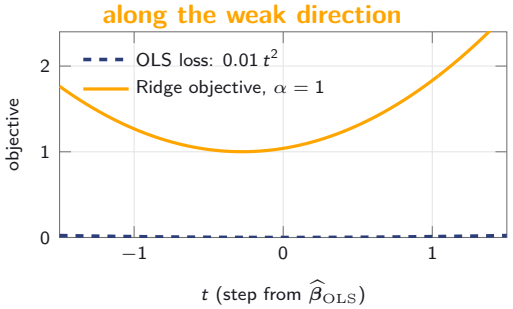
The rows are now $21\beta_1 + 20\beta_2 = 40$ and $20\beta_1 + 21.04\beta_2 = 40.048$.

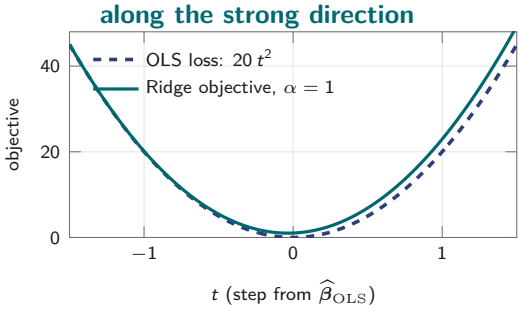
	OLS	Ridge
determinant	0.8	41.84
small eigenvalue	0.020	1.020
line slopes	-1.000, -0.998	-1.050, -0.951

Insight

The two equations now differ in slope by 0.10 instead of 0.002 — fifty times more. A crossing of two lines that visibly cross is a stable thing.

Ridge Lifts The Valley Floor





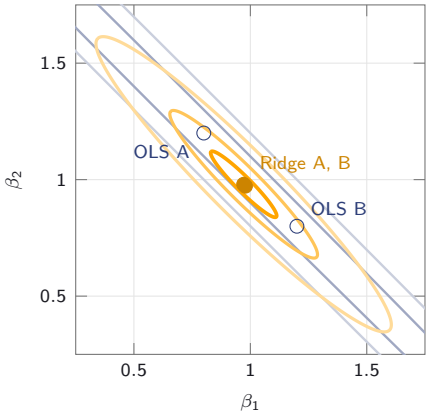
Adding αI adds the same curvature α to *every* direction. Relative to what was there, that is not the same at all:

$$\text{weak: } 0.02 \to 1.02 \ (\times 51), \qquad \text{strong: } 40.02 \to 41.02 \ (\times 1.025).$$

Insight

Ridge barely touches the direction the data already determine. It rebuilds the one they do not.

Ridge Fills The Valley



Blue: OLS loss contours from before. Orange: the same levels of the Ridge objective with $\alpha = 1$.

The needle has become a rounded bowl. Its floor is no longer flat, so there is one lowest point, and a small change in the responses moves it only a little.

The two OLS minima were 0.566 apart. The two Ridge minima are 0.011 apart — they sit on top of each other at this scale.

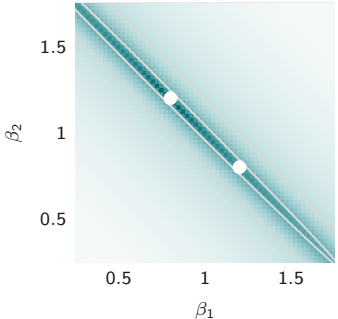
Summary

Same data, same noise. The difference is the shape of the surface being minimised.

Interactive explorer: raise α and watch the valley close ↗

Two Maps: The OLS Valley And The Ridge Bowl

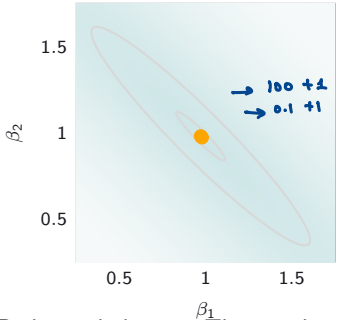
OLS: $\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2$



Left: a dark band. Every point on it is nearly as good as every other; the two white dots are 0.57 apart and the loss cannot separate them.

Ridge does not move the walls. It tilts the floor.

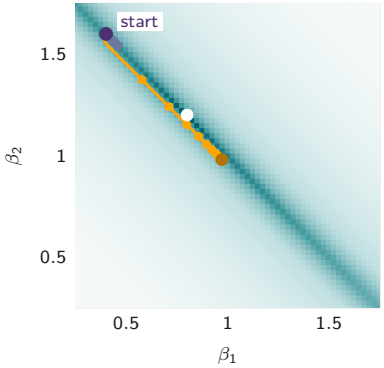
Ridge, $\alpha = 1$: the same plus $\frac{1}{2}\|\boldsymbol{\beta}\|^2$

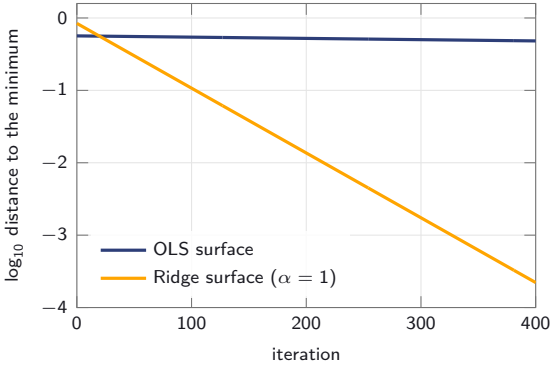


Right: a dark *spot*. The penalty raises the far ends of the band, so the surface has one bottom; the two orange dots (Ridge on A, Ridge on B) are 0.011 apart.

Interactive explorer: raise α and watch the band become a spot ↗

Gradient Descent On Each Map: A Crawl And A Walk





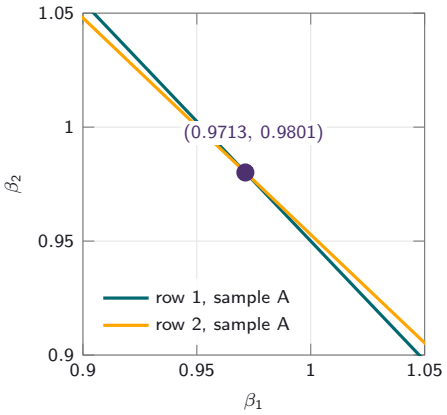
Same start, same step $\eta = 0.02$, 400 steps. Each step multiplies the error along an eigenvector by $1 - \eta\mu$:

$$\text{along the floor: OLS } 1 - 0.02 \times 0.02 = 0.9996, \qquad \text{Ridge } 1 - 0.02 \times 1.02 = 0.9796.$$

Blue reaches the band in a few steps and then crawls: still 0.48 away after 400. Orange walks to the bottom: 2×10^{-4} away.

Interactive explorer: choose a start and a step size ↗

The Ridge Coefficients For Sample A



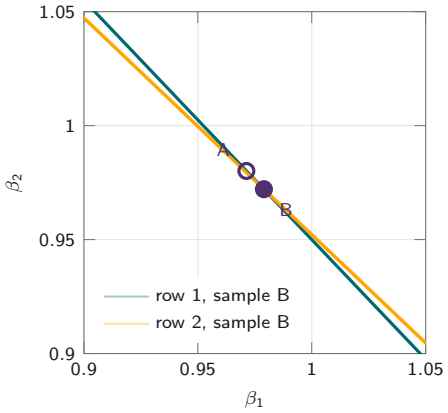
Using the original responses and $\alpha = 1$,

$$\widehat{\beta}^{(A)}_{\text{Ridge}} = \frac{1}{41.84}\begin{bmatrix} 21.04 & -20 \\ -20 & 21 \end{bmatrix}\begin{bmatrix} 40 \\ 40.048 \end{bmatrix} = \boxed{\begin{bmatrix} 0.9713 \\ 0.9801 \end{bmatrix}}.$$

	$\widehat{\beta}_1$	$\widehat{\beta}_2$
OLS	0.8000	1.2000
Ridge, $\alpha = 1$	0.9713	0.9801

The window is 0.15 wide — zoomed in eight times relative to the OLS slides.

The Ridge Coefficients For Sample B



Same $\alpha = 1$, the perturbed responses:

$$\widehat{\boldsymbol{\beta}}^{(B)}_{\text{Ridge}} = \frac{1}{41.84}\begin{bmatrix} 21.04 & -20 \\ -20 & 21 \end{bmatrix}\begin{bmatrix} 40 \\ 40.032 \end{bmatrix} = \boxed{\begin{bmatrix} 0.9790 \\ 0.9721 \end{bmatrix}}.$$

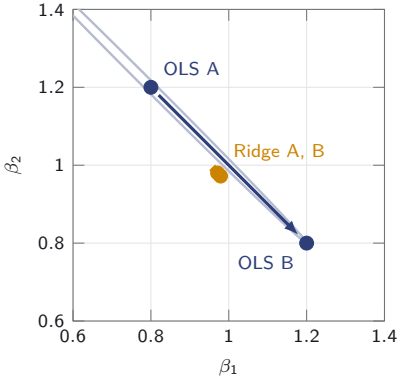
Row 1 does not move at all ($x_1^\top \boldsymbol{y}$ is 40 in both samples); row 2 shifts by a hair; the crossing moves

$$\|\widehat{\boldsymbol{\beta}}^{(B)}_{\text{Ridge}} - \widehat{\boldsymbol{\beta}}^{(A)}_{\text{Ridge}}\|_2 \approx 0.0111.$$

Summary

The same perturbation moved the OLS crossing by 0.566 and the Ridge crossing by 0.0111.

The Full OLS And Ridge Comparison



Method	Sample A $\widehat{\beta}_1$	Sample A $\widehat{\beta}_2$	Sample B $\widehat{\beta}_1$	Sample B $\widehat{\beta}_2$
OLS	0.8000	1.2000	1.2000	0.8000
Ridge, $\alpha = 1$	0.9713	0.9801	0.9790	0.9721

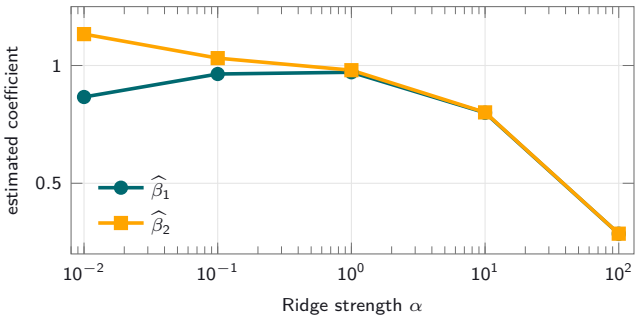
Training fit

OLS: zero training loss in both samples. Ridge: a small error, because the penalty also counts.

Stability

Blue arrow: the OLS jump. Orange arrow: the Ridge jump, too short to see. Same noise, fifty-fold difference.

Ridge As The Penalty Strength Changes



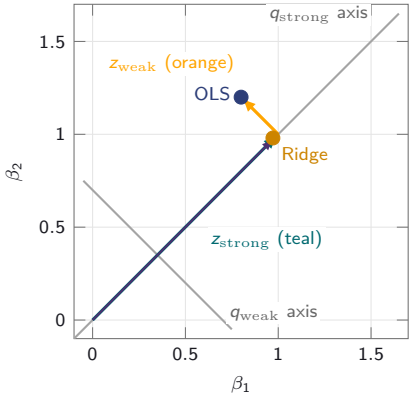
Near $\alpha = 0$, the estimate is close to OLS. Moderate α suppresses the unstable difference between the two coefficients. Very large α shrinks both coefficients strongly toward zero.

Warning

Training error alone always prefers $\alpha = 0$. Validation data must choose the useful amount of shrinkage.

Interactive explorer: trace the Ridge and Lasso paths ↗

OLS And Ridge In The Eigenvector Basis



Write $\boldsymbol{X}^\top \boldsymbol{X} = Q\,\mathrm{diag}(\mu_1, \mu_2) Q^\top$ and $c_j = q_j^\top \boldsymbol{X}^\top \boldsymbol{y}$. Each method estimates one coordinate per eigenvector:

$$\widehat{z}_j^{\,\mathrm{OLS}} = \frac{c_j}{\mu_j}, \qquad \widehat{z}_j^{\,\mathrm{Ridge}} = \frac{c_j}{\mu_j + \alpha} = \frac{\mu_j}{\mu_j + \alpha}\,\widehat{z}_j^{\,\mathrm{OLS}}.$$

In the picture, the OLS estimate is 1.414 along the strong axis plus 0.282 along the weak one. Ridge keeps 1.380 of the first and 0.006 of the second — it slides the point almost onto the strong axis.

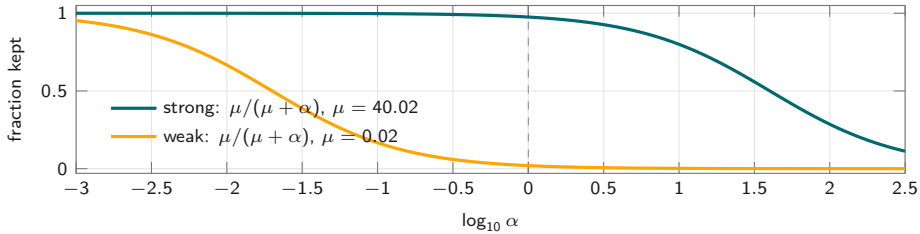
Insight

Ridge shrinks each direction by how strongly the data support it.

Directional Shrinkage In The Numerical Example

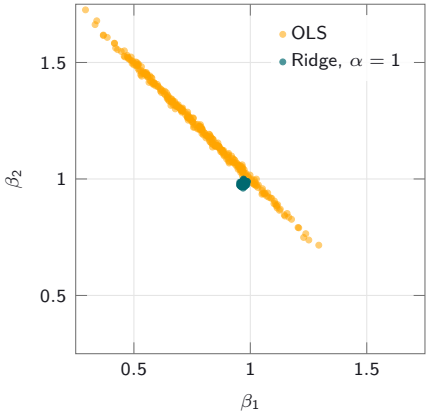
For $\alpha = 1$:

Direction	μ_j	shrinkage $\mu_j/(\mu_j+1)$	$\widehat{z}_j^{\mathrm{OLS}}$	$\widehat{z}_j^{\mathrm{Ridge}}$
weak	0.01999	0.0196	0.2821	0.0055
strong	40.0200	0.9756	1.4144	1.3799



The weak coordinate is gone by $\alpha = 0.1$, long before the strong one is touched. The window where the first has happened and the second has not is what a validation search finds (Step 2).

Three Hundred Training Samples, Two Estimators



Each dot is one complete fit: the four responses were re-drawn with noise of size 0.04 — the same size as the A-to-B change — and the model refitted. Three hundred times.

spread (s.d.)	OLS	Ridge
along the floor	0.288	0.006
across it	0.007	0.007

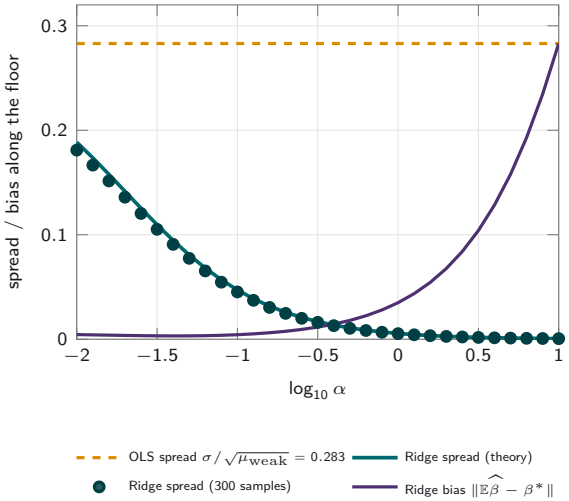
OLS wanders **43 times** further along the floor than across it. Ridge cuts the along-floor spread **51-fold** and leaves the other alone.

Summary

A tighter cloud is lower estimator variance. This is what "stable" means.

Interactive explorer: draw your own three hundred samples ↗

How Much Ridge Narrows The Cloud, And What It Charges



The cloud on the previous slide, measured along the floor:

$$\text{OLS: } \frac{\sigma}{\sqrt{\mu_{\text{weak}}}} = \frac{0.04}{0.141} = 0.283,$$

$$\text{Ridge: } \frac{\sigma\sqrt{\mu_{\text{weak}}}}{\mu_{\text{weak}}+\alpha} = 0.0055 \quad (\alpha = 1).$$

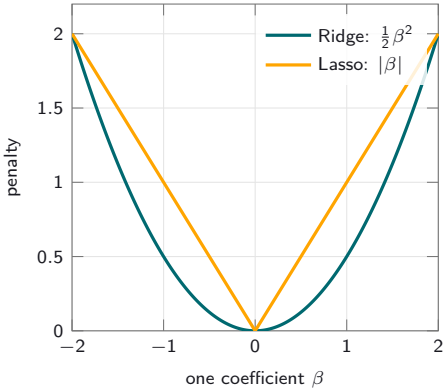
Dots: simulation. Line: the formula. Purple: the price, the mean estimate drifting from $\beta^* = (1, 1)$ toward the origin.

Insight

Variance falls fifty-fold before bias reaches 0.04. That trade is why Ridge wins here.

Interactive explorer: draw your own samples at any α ↗

Lasso Changes The Penalty



Method	Objective	Behaviour
Ridge	$\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \frac{\alpha}{2}\sum_j \beta_j^2$	smooth shrinkage
Lasso	$\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \alpha\sum_j \|\beta_j\|$	shrinkage, exact zeros

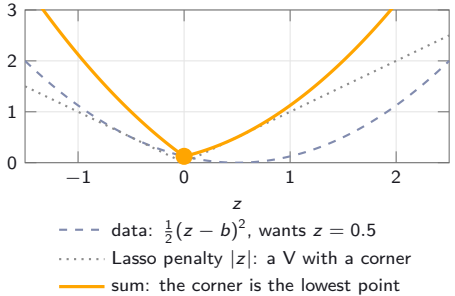
Insight

Near zero the Ridge penalty is flat — its slope β vanishes, so it never quite pushes a coefficient to zero. The Lasso penalty keeps slope ± 1 all the way in: a corner, and an optimum can sit exactly on it.

Lasso Is Friction, Ridge Is A Spring

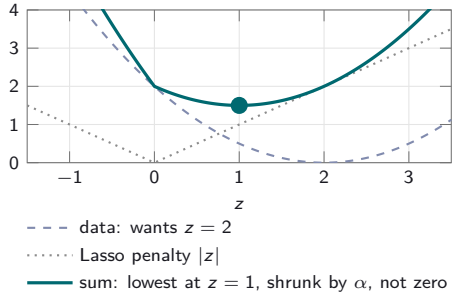
One coefficient z. The data pull it toward b with the bowl $\frac{1}{2}(z-b)^2$. The penalty pushes back. Add the two curves and look at what happens at $z = 0$.

weak pull, $b = 0.5$: the corner holds



Forces at $z = 0$. The data pull with strength $|b|$. Ridge's penalty $\frac{\alpha}{2}z^2$ is a *spring*: its push αz is *zero* at the origin, so any pull at all moves the coefficient.

strong pull, $b = 2$: the corner gives way

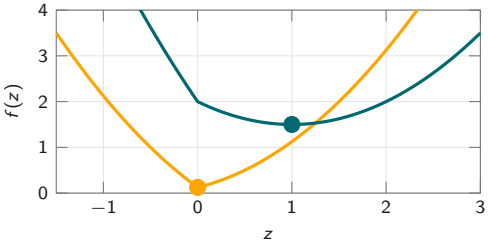


Lasso's penalty $\alpha|z|$ is *friction*: a constant push of α against leaving 0 in either direction. Pull weaker than friction, $|b| \leq \alpha$: **the coefficient does not move**, exactly zero. Stronger: it slides until the leftover pull equals the friction, $z = b - \alpha$.

The Same Thing In Slopes: Where Does f' Cross Zero?

One coefficient z; the data want $z = b$; the objective is $f(z) = \frac{1}{2}(z-b)^2 + \alpha|z|$ with $\alpha = 1$. A minimum is where the slope f' crosses zero. The penalty $|z|$ has slope $+\alpha$ to the right of 0 and $-\alpha$ to the left, so f' **jumps by** 2α **at** $z = 0$.

the objective

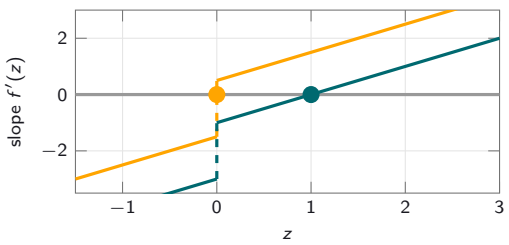


$b = 0.5 \le \alpha$: minimum at $z = 0$

$b = 2 > \alpha$: minimum at $z = 1$

If $|b| \le \alpha$ the slope is negative just left of the corner and positive just right of it: the corner is the minimum. Otherwise the crossing is at $z = b \mp \alpha$: shrunk by α, not zero.

its slope: where does it cross zero?



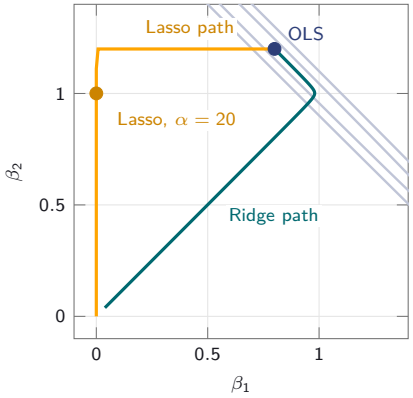
$b = 0.5$: the jump straddles 0, so $z = 0$ wins

$b = 2$: the jump misses 0; crossing at $z = 1$

$$\widehat{z}_{\text{Lasso}} = \text{sign}(b)\max(|b| - \alpha,\, 0).$$

Ridge cannot do this: $\frac{\alpha}{2}z^2$ has slope $\alpha z = 0$ at the origin, no jump, so the crossing is at $b/(1+\alpha)$: small, never zero.

Where The Ellipses And Coefficient Constraints Come From



Around $\widehat{\beta}_{\mathrm{OLS}}$ the loss is exactly $J_{\mathrm{OLS}}(\beta) = J^* + \frac{1}{2}(\beta - \widehat{\beta}_{\mathrm{OLS}})^\top \boldsymbol{X}^\top \boldsymbol{X}(\beta - \widehat{\beta}_{\mathrm{OLS}})$, whose level sets are the blue ellipses. A penalty is the same as a budget:

$$\text{Ridge: } \min J_{\mathrm{OLS}} \text{ subject to } \beta_1^2 + \beta_2^2 \le s^2,$$
$$\text{Lasso: } \min J_{\mathrm{OLS}} \text{ subject to } |\beta_1| + |\beta_2| \le s.$$

Each α corresponds to one budget s, and as s shrinks the solution traces a path from the OLS point toward the origin — teal for Ridge, orange for Lasso.

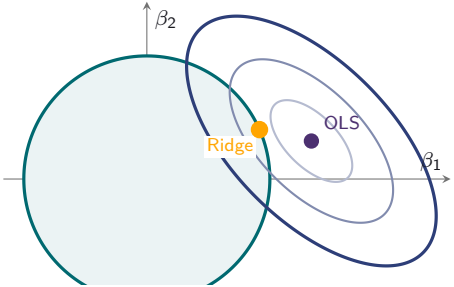
Summary

Ridge slides down the trough floor to $(1, 1)$ and then toward the origin. Lasso runs along $\beta_2 \approx 1.2$, hits the axis near $\alpha = 17$, and only then shrinks β_2.

Loss Ellipses Meet The Ridge Circle And Lasso Diamond

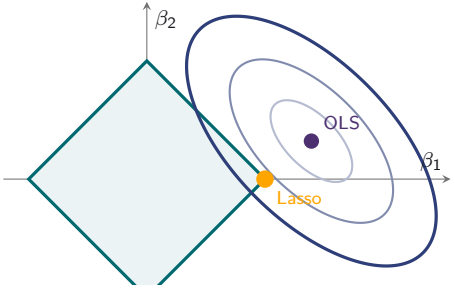
A *schematic* pair of moderately correlated predictors. (The running example's ellipses are 45:1 and would draw as a single line.) Blue curves: equal OLS loss. Orange point: first contact with the allowed region.

Ridge budget: $\beta_1^2 + \beta_2^2 \le s^2$



Smooth contact: both coefficients stay nonzero.

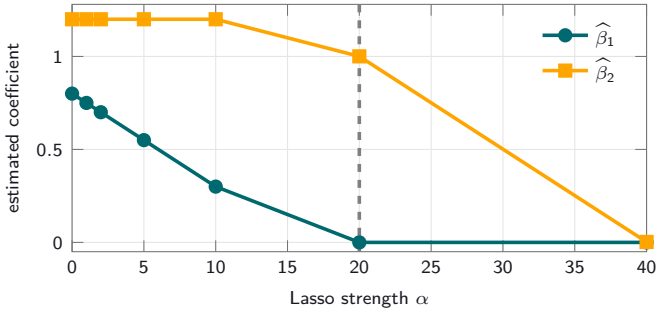
Lasso budget: $|\beta_1| + |\beta_2| \le s$



A corner lies on an axis: one coefficient is exactly zero.

Interactive explorer: slide the constraint from diamond to circle ↗

Lasso On The Same Numerical Example



At $\alpha = 0$, Lasso equals OLS:

$$(0.8, 1.2).$$

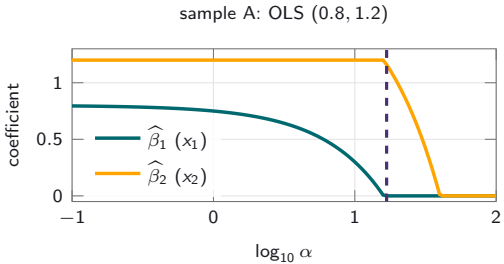
As α increases, one coefficient reaches exactly zero:

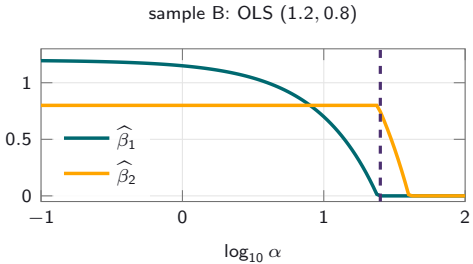
$$\alpha = 20: \qquad \widehat{\beta}_{\text{Lasso}} \approx \begin{bmatrix} 0 \\ 1.0004 \end{bmatrix}.$$

Warning

Lasso keeps one of two near-duplicate predictors and drops the other. Which one survives is decided by a margin of 0.1% (next slide): a different noise draw or different units can change it, while the prediction stays useful.

Lasso Keeps One Twin. Which One Is Decided By 0.03





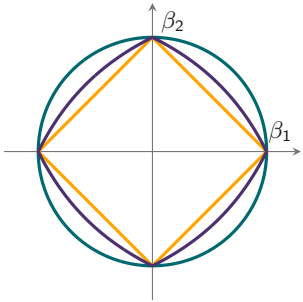
Both samples: $\widehat{\beta}_1 \to 0$ exactly ($\alpha = 16.8$, then 25.1) and x_2 carries the sum, because $\boldsymbol{X}^\top \boldsymbol{y}_A = (40.000, 40.048)$, $\boldsymbol{X}^\top \boldsymbol{y}_B = (40.000, 40.032)$: x_2 wins by 0.1%, its column being 0.2% longer.

Warning

Ridge split the credit 0.97/0.98; Lasso awards it all to the twin that wins by a rounding error. Between near-duplicates, a Lasso zero reports units and noise, not the world.

Interactive explorer: switch samples and watch which coefficient reaches zero ↗

OLS, Ridge, Lasso, And Elastic Net



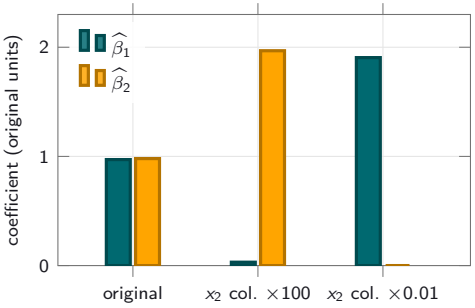
Method	Objective	With correlated predictors
OLS	$\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$	coefficients can be unstable
Ridge	OLS loss $+\frac{\alpha}{2}\|\boldsymbol{\beta}\|_2^2$	shares weight between them
Lasso	OLS loss $+\alpha\|\boldsymbol{\beta}\|_1$	may keep one representative
Elastic Net	OLS loss $+\alpha_1\|\boldsymbol{\beta}\|_1 + \frac{\alpha_2}{2}\|\boldsymbol{\beta}\|_2^2$	sparsity plus stability; can keep a correlated group

Summary

The allowed region's *corners* are what produce exact zeros; its *curvature* is what shares weight. Elastic Net keeps both. Standardize the predictors first, and leave the intercept unpenalized.

Step 0: Why The Units Of A Feature Change The Answer

Record x_2 in different units — multiply the column by 100, as in centimetres instead of metres. Nothing about the data changes. Fit Ridge with the same $\alpha = 1$ and convert back:



x_2 recorded in	$\widehat{\beta}_1$	$\widehat{\beta}_2$
original units	0.971	0.980
units 100$\times$ smaller	0.031	1.968
units 100$\times$ larger	1.905	0.000

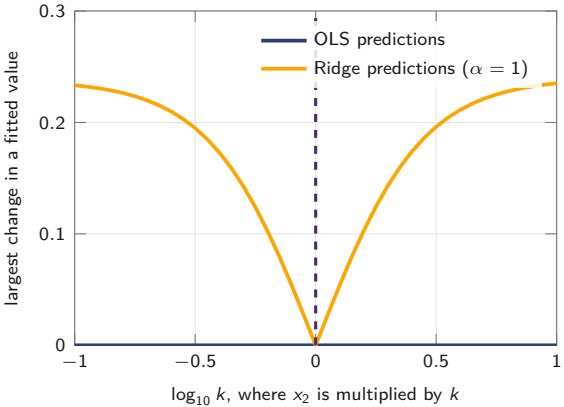
The penalty treats β_1 and β_2 as the same kind of number. A column 100$\times$ bigger gets a coefficient 100$\times$ smaller, which the penalty barely notices — so it collects all the credit.

Warning

Which feature "matters" was decided by a choice of units. So the scales must agree first.

Interactive explorer: rescale one feature and compare OLS with Ridge ↗

Rescale One Feature: OLS Does Not Notice, Ridge Does



Multiply the second column by k and refit. OLS returns $\widehat{\beta}_2/k$ and *every fitted value is identical*: the blue line is zero to machine precision.

Ridge's fitted values move by up to 0.23 in either direction, because the penalty charges β_2^2 and β_2 has changed size while the data have not.

Warning

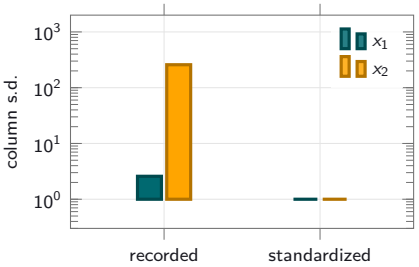
A penalty is a statement about the size of coefficients. It means nothing until every coefficient is measured in the same unit. That is the whole reason Step 1 exists.

Interactive explorer: slide k and watch only the orange points move ↗

Step 1: Standardize — Using Training Statistics Only

For each feature j, compute the mean and standard deviation *on the training rows*, then apply those same two numbers everywhere:

$$z_{ij} = \frac{x_{ij} - m_j}{s_j}, \qquad m_j,\ s_j \text{ from training rows only.}$$



Left: the $\times 100$ recording, before and after. Running example ($m_j = 0$ already): $s_1 = 2.582$, $s_2 = 2.585$; after scaling $\widehat{\beta}_{\text{Ridge}} = (2.213,\ 2.216)$ — equal credit, whatever the units were.

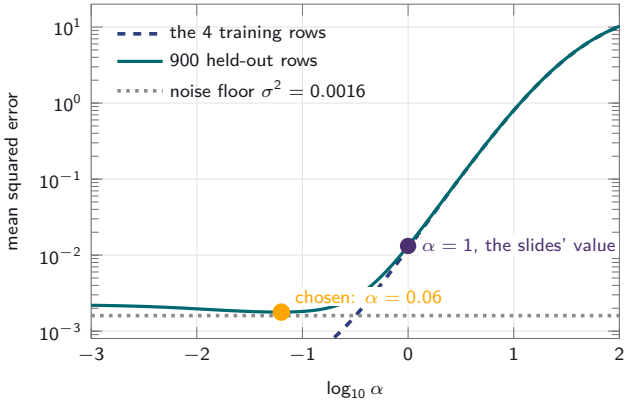
What to store

m_j, s_j are part of the model; new observations are scaled with them.

Warning

Computing m_j, s_j on all rows, test included, leaks information from the rows you will later score on.

Step 2: Choose α By Validation Error, Never By Training Error



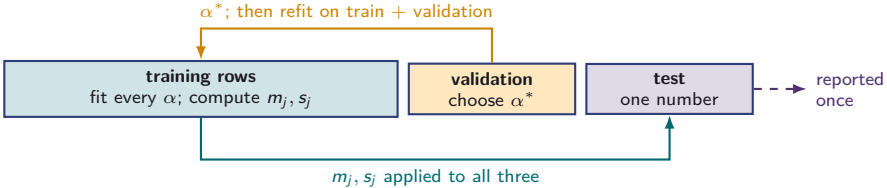
Same setting as the whole lecture: sample A's four rows are the training set; 900 fresh rows drawn the same way ($x_2 \approx x_1$, noise 0.04) are the validation set. Ridge is fitted on the four rows for every α and scored on the 900.

Training error only rises with α: OLS already minimizes it, so it always votes for $\alpha = 0$. Validation error falls from 0.0022 (OLS) to 0.0018 at $\alpha \approx 0.06$, nearly the noise floor, then climbs.

Warning

$\alpha = 1$ was chosen to make the pictures legible. On new rows it costs 0.013: it kills the weak direction, which helps, but shrinks the strong one by 2.4%, which hurts. Validation finds the value that does the first without the second.

Step 3: Evaluate Once On Test Data, Then Stop



Why a third set

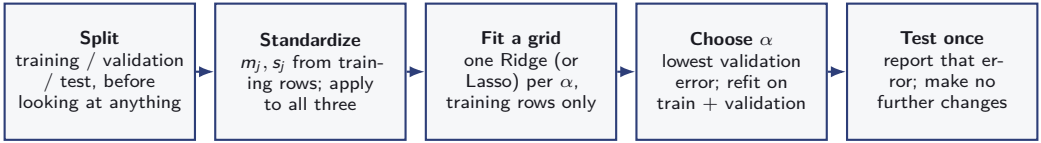
Validation error was *used* to pick α, so it is now optimistic — you chose the value on which it looked best. The test rows never influenced any decision, so their error is honest.

Warning

"Once" is literal. Peek, adjust, look again, and the test set has quietly become a second validation set.

What you ship: $\underbrace{(m_j, s_j)}_{\text{training scaling}}$ $\underbrace{\alpha^*}_{\text{chosen by validation}}$ $\underbrace{\widehat{\beta}}_{\text{refitted on train + validation}}$ $\underbrace{\widehat{\text{err}}_{\text{test}}}_{\text{reported once}}$ — plus the intercept, fitted and never penalized.

The Complete Modeling Procedure



Warning

The four ways this goes wrong. Scaling with all rows (leak). Choosing α by training error (always 0). Penalizing the intercept (shifts every prediction). Reading the test error more than once (it stops being a test).

Summary

Regularization sacrifices a little training fit for a model whose coefficients and predictions depend less on the particular sample. The procedure above is how that trade is made honestly — every choice is scored on rows that did not shape it.

Open the OLS, Ridge and Lasso explorer ↗